

FVCOM-MP: Late-Time Tracer Diagnostics

Delaware River Estuary Test Cases

WaterPACT FVCOM-MP Development

May 2026

Current FVCOM-MP test outputs now span multi-month integrations (> 3 months), enabling day-30 and day-90 tracer checks beyond the initial 10-day mass-balance archive.

Delaware River Test Cases

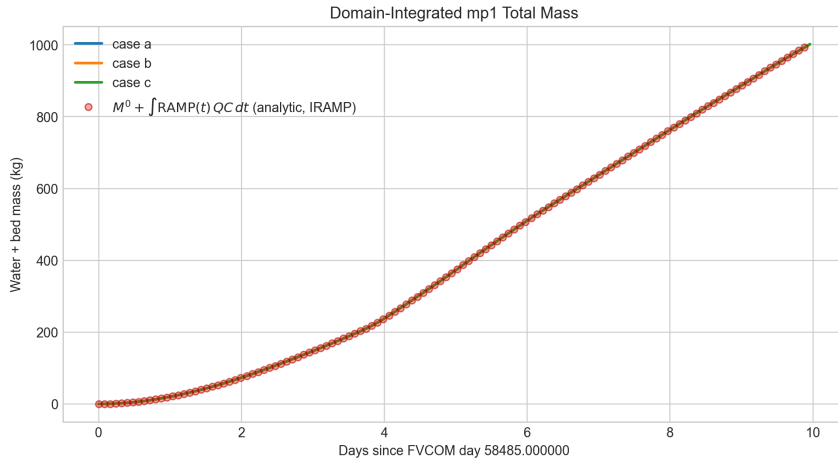
	Case a	Case b	Case c
Role	<i>baseline (published)</i>	sensitivity b	sensitivity c
PLAST_FLOC	F	T	T
SEDIMENT_MODEL	F [†]	T	T
SED_SD50	—	0.1 mm	1.0 mm
MP water field(s)	c	$c_a + c_d$	
Agg. gate $d_{MP}^{\max}$	—	34 μm (excl.)	162 μm (act.)
		$c_d \approx -0.39 \text{ mm s}^{-1}$	$c_a = +3.0 \text{ mm s}^{-1}$
MP settling est.	—	buoyant/free MP	$\Delta \approx 3.4 \text{ mm s}^{-1}$

c = disaggregated conc.

c_a = aggregated, c_d = disaggregated (floc path)

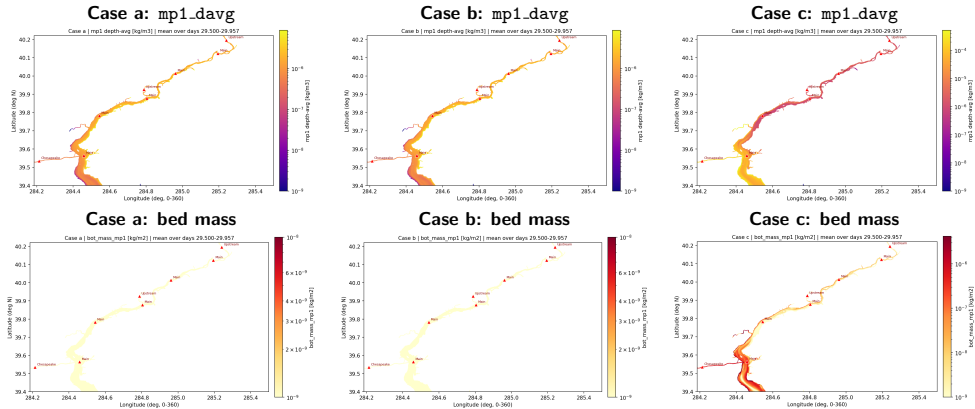
[†]sed. initialized; adv_scal/FCT bypassed • SOURCE_AG_RATE = 0 $\Rightarrow c_a \equiv 0$ • $\rho_p = 950 \text{ kg m}^{-3}$, $d_p = 100 \mu\text{m}$ • settling estimate uses $\rho_w \approx 1025 \text{ kg m}^{-3}$ and SED_WSET = 3 mm s⁻¹

First 10-Day Mass Balance: v012 Ramp-Corrected Analytic Archive

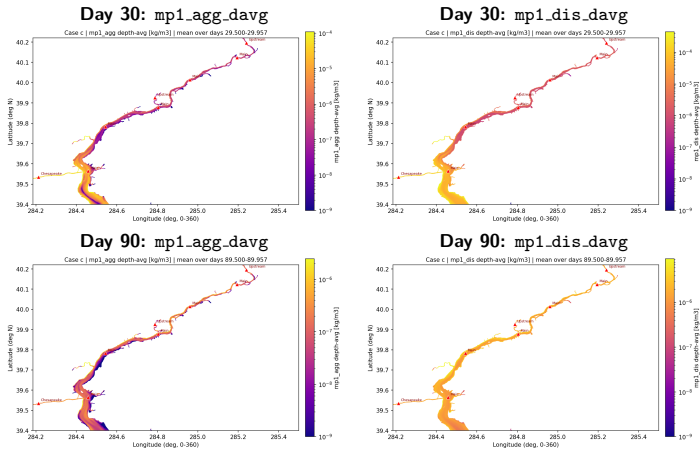


Final 10-day archive check: total mass by case is well behaved after the ramp-corrected analytic source treatment.

Day 30 Spatial Contrast: Water Column and Bed Mass



Floc-Component Split: Case c, Day 30 vs. Day 90



Case **c** shows the active floc pathway; case **b** remains effectively zero in the aggregated component under the fine-sediment parameterization.